

What did I learn?

I learned how classes and objects actually work — before this project I didn't really get the difference between a class and an object, but building Assignment, Homework, and Exam made it click. I learned that a class is just a blueprint, and you have to actually create an object from it before it does anything. I also learned inheritance and `super()` — how Homework and Exam could reuse Assignment's setup instead of repeating the same code twice. I understood why `self` only works inside a class, not in a normal function, after mixing that up a few times. I also got a lot more comfortable reading actual Python error messages instead of guessing — things like `TypeError` and `AttributeError` started actually telling me what was wrong instead of just being scary red text.

Challenges I faced

I spent several days on this project, working through it step by step over multiple sessions. Some of the hardest moments:

I tried to create an assignment using a tuple, `Assignment = (date, subjects, score)`, and then tried to add attributes to it like `Assignment.date = ...` errors everywhere

I kept mixing up indentation, especially with functions inside classes — I'd write a method at the wrong indentation level and it would either break the class or run outside it completely. Especially after the menu and with the `(if/elif/else)`

I put my empty-list checks (`if self.assignments == []`) inside for loops, not realizing that if the list is empty, the loop body never runs at all — so the check never actually fired. I made this same mistake twice, once in `list_assignments` and again in my filter code.

My subject filter didn't work when someone typed a subject with different capitalization (like "english" instead of "English") — I didn't know Python string comparison is case-sensitive until I actually tested it and saw it fail. I heard Navin say it though but it made sense doing it. I created objects like `Homework()` or `Exam()` without passing in the required arguments, which caused `TypeError`s until I understood that a constructor needs all its required values immediately when the object is created.

I struggled with getting my menu to actually loop — I tried making the function call itself again instead of using a proper `while True:` loop, which caused the tracker to reset every time.

How I'd improve it, given more time. I would genuinely try to spend more time coding as I have seen that the more I practise, the better I become and It is hard. I don't believe Navin when he says it isn't.